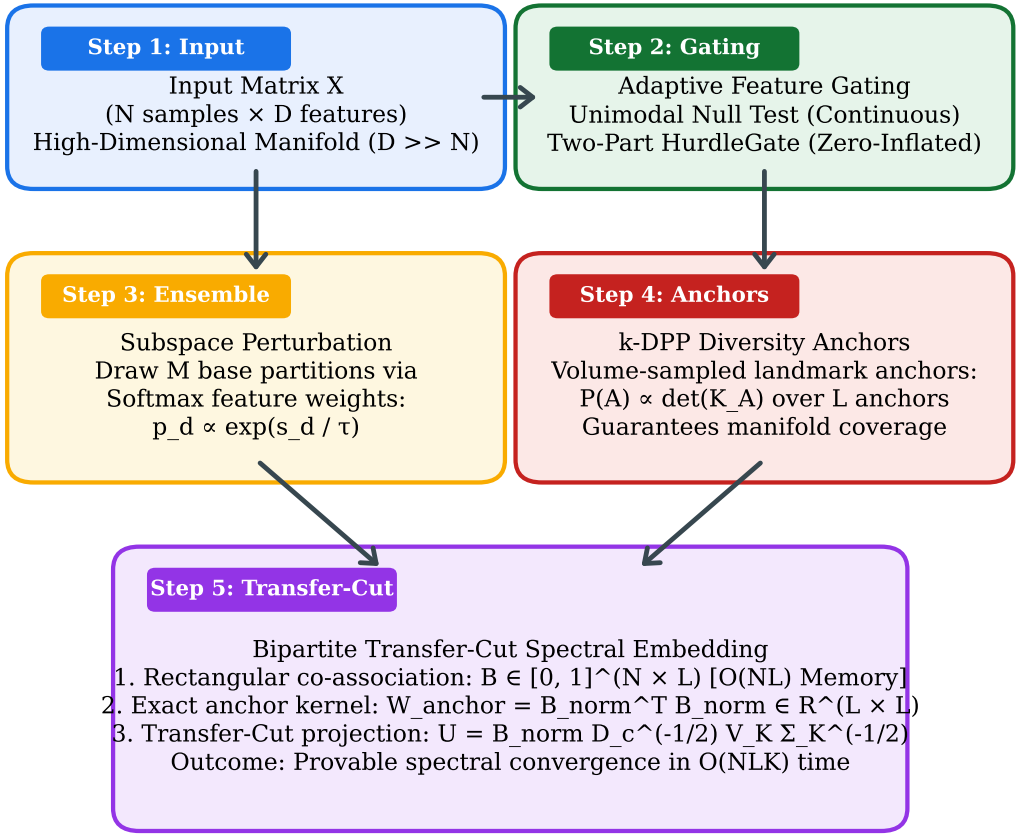
(a) NullConsensus Bipartite Landmark Consensus Architecture



(b) Zero-Inflation Dropout Collapse vs. HurdleGate Solution

PATHOLOGY: Marginal Gating Collapse

- Single-cell RNA-seq exhibits 60%–95% technical dropout zeros.
- Massive Dirac mass at zero (δ_0) distorts higher standardized moments:
  Excess Kurtosis: $\kappa_d \to \infty$, Skewness: $\gamma_d \to \infty$
- Pearson Bimodality Coefficient evaluates to artificial ceiling:
  $BC_d = (\gamma_d^2 + 1) / (\kappa_d + 3) \longrightarrow 1.0$ (Artificial Bimodality Spike)
- Outcome: Standard gating selects 100% noise genes with extreme dropout, triggering catastrophic clustering failure (ARI ≈ 0 in Paper 1).

SOLUTION: Two-Part HurdleGate Gating

Decouples zero-inflation from positive-expression multimodality:

$$s_{Hurdle}(d) = \alpha \cdot Entropy(p_{0,d}) + (1 - \alpha) \cdot BC(x_d^+)$$

1. Bernoulli Component: $p_{0,d} = (1/N) \sum 1\{x_{id} = 0\}$
   Measures informative lineage dropout frequency via Bernoulli entropy.
2. Continuous Support Component: $x_d^+ = \{x_{id} \mid x_{id} > 0\}$
   Evaluates true biological multimodality strictly on expressing cells.

Outcome: Eliminates Dirac artifact; restores +0.3832 ARI Recovery.